

MANUAL J ASSESSMENT

28 April 2026

(per site visit on 4/13/26 and plan '7/1/16' provided)

CLIENT/BUILDER:

Michael and Raquel Wachter

PLAN/ADDRESS:

4308 Vivian Street

SYSTEM BREAKDOWN

AHU/SYS.ID	AHRI#	Tonnage	Zoned (Y/N)	Cooling Sensible / Latent Btuh	Equip/ Load %	SEER	S/A Duct Configuration	F/A
AHU-1 1 st Floor less stairway	9169630	*2.5 Tons **2.0 Tons	N	S: 15,803 L: 2,706	-	15.50	Trunk & branch, perimeter outlets	***24 CFM
AHU-2 2 nd Floor w/ stairway	203474660	*3.0 Tons **2.5 Tons	N	S: 20,954 L: 5,137	-	15.50	Trunk & branch, perimeter outlets	***60 CFM

*As-installed – unable to verify via Manual S; expanded data not available via Carrier's Performance Calculator portal

**Estimated based on total cooling load

***Fresh air ventilation not installed when inspected – required – split between systems as needed shown (can be merged through one system)

SPECIFICATIONS

Design Condition Input: **100/75** (Houston Dunn, TX. @ 0.4% Clg. DB – ASHRAE 2021)

Internal Gains: ACCA Table 6A, 6B / Manual J 8th Ed.

Fresh-air: None visible, although required (shown above per **ASHRAE 62.2-2010**)

Orientation: per 'Site Plan', verified via site visit

Infiltration Const. Quality / Infiltration Method: 'Average' (assumed – verify via blower door test) / 'Simplified'

Duct Configuration: **R-8** (S/A) & **R-8** (R/A); Return as 'radial'

Equipment Manufacturer: **Carrier**

- Furnace Efficiency: **96%**
- Coil/Air Handler Configuration: **multi-poise evap. coil**
- Condenser/Heat Pump: **Electric A/C**

-Insulation per Attic Insulation Chart-

Wall insulation: **R-13**

Cold floor insulation: **R-13 @ sealed/tight crawlspace, R-20 @ ambient floor, R-13 over Garage**

Attic/Ceiling Insulation: **R-21 @ rafters**

Glazing NFRC: **N/A – varies, & w/ internal shading where visible per site visit report**

DISCLAIMER

Specifications assumed for the purpose of providing an accurate representation of each home's heat loss/heat gain and subject to change overall sizing in the chance that specifications differ from the above mentioned.